

SHUVAM PAL

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Education

Dayananda Sagar College of Engineering

Bengaluru, India

B.E. Artificial Intelligence and Machine Learning

Experience

Bentham AI

Dec 2025 – Feb 2026

Backend Engineering Intern: Go, Node.js, Docker, GCP, Puppeteer, GitHub Actions

Remote

- Built a resilient **Puppeteer/Node.js workflow** for MCA filings with session handling, CAPTCHA handoff, form retries, and state recovery, reducing manual filing time by **85%**.
- Optimized an **AI-assisted company-registration pipeline** for unique company-name suggestions, descriptions, and filing-field drafts, reducing manual drafting/rework across MCA workflow preparation.
- Containerized backend services with **Docker**, wired **GitHub Actions** deployments to **Google Cloud Run**, and improved release velocity by **70%**.

Google Summer of Code

May 2025 – Sept 2025

Mentee at Invesalious: Python, VTK, wxPython, NumPy, PyTorch, ONNX

Remote

- Built Python medical-image segmentation tooling for **95 brain subparts**, handling model inference, preprocessing, orientation correction, and large-volume data paths with reproducible outputs.
- Reduced segmentation runtime by **87%** using asynchronous and parallel execution while preserving inspection workflows for generated masks and 3D surfaces.
- Integrated production-facing changes in a mature open-source codebase with review constraints, compatibility concerns, and careful debugging across existing workflows.

Open Source

[Contributions](#)

- Contributed patches across Invesalious, IOOS, VideoLAN, CRIU, Tiled, Kyverno, and Kuadrant, working inside maintainer review processes for reliability, security, and integration quality.

Projects

Seaweed: Contest Platform | *Go, React, Vite, PostgreSQL, Judge0, Firebase, Redis, AWS, Docker, Kubernetes* [GitHub](#)

- Built a full-stack **Contest Platform** with React/Vite and Go for registration, submissions, automated judging, student-facing scoreboards, and live progress/ranking views across **500+ concurrent users**.
- Designed async Judge0 dispatch, PostgreSQL materialized-view leaderboards, Redis runtime state, and Docker/Kubernetes deployment on AWS, achieving **sub-5s P95 verdict latency** and **70%** lower infrastructure cost.

Postificus | *Go, Echo, Rod, PostgreSQL, Redis, RabbitMQ, Docker, AWS, Prometheus*

[GitHub](#)

- Built Go/Echo backend services for an **AI-assisted content distribution platform** using browser automation, with publishing workflows, editor autosave, RabbitMQ workers, and PostgreSQL persistence.
- Added DLQs, retries, circuit-breaker isolation, AWS storage, health checks, and Prometheus telemetry so unreliable external API/browser workflows stayed recoverable across distributed workers.

Murdoc | *Python, FastAPI, OpenTelemetry, Prometheus, Docker, MCP*

[GitHub](#)

- Built a self-hosted **AI security gateway** for LLM, retrieval, HTTP-tool, and MCP traffic with request scoring, allow/block policies, PII redaction, and audit logs before downstream execution.
- Blocked **90%+** prompt-injection and unsafe-tool attempts across a **150-case attack corpus** with false positives under **5%**; added Prometheus/OpenTelemetry observability for replayable decisions.

Technical Skills

Languages: Go, Python, JavaScript/TypeScript, C++

Frameworks: React, Vite, Tailwind CSS, FastAPI, Echo, Node.js, GraphQL

Tools: PostgreSQL, Redis, RabbitMQ, Docker, Kubernetes, AWS, GCP, Firebase, Git, GitHub Actions, Prometheus, OpenTelemetry, Puppeteer, Go-Rod